

23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

2. UI Design Document

Group No : C13

Team Members :

Roll No	Student Name
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Project Title : Campus Canteen Pre-Order and Queue Management System

2.1. Functional requirements:

[list the main functionalities required]

1. Students can browse the daily menu (with category, price, prep time, and live stock) and place pre-orders for a selected pickup slot.
2. The system calculates and displays an estimated waiting time based on pending orders and kitchen capacity.
3. Order status updates automatically through defined stages — Placed → Preparing → Ready → Collected — with each change logged (via a database trigger) to OrderStatusLog.
4. Students can track their live order status and view past order history.
5. Canteen staff can view a live queue of all incoming orders and update order status from a dashboard.
6. Preorders are prioritized ahead of walk-in orders based on booking time/pickup slot.
7. Admin can manage menu items, pricing, and daily stock; items go out of stock automatically at zero.
8. Admin can view reports — peak ordering hours, most popular items, daily sales, average preparation time.
9. Role-based access: students see only their own orders; staff see/update all active orders; admin manages menu and system-wide data.
10. Order placement is transaction-safe — if an item goes unavailable mid-order, the transaction rolls back.

2.2. Non functional requirements:

[list the main qualitative features required]

1. **Usability** – simple, intuitive interface for students, staff, and admin with minimal training needed.
2. **Responsiveness** – UI adapts cleanly to desktop and mobile screens (built with Bootstrap's responsive grid).
3. **Performance** – order status and queue updates should reflect in near real time.
4. **Reliability/Data Integrity** – transaction management and rollback ensure no inconsistent order states.
5. **Security** – role-based access control restricts each user type to appropriate data/actions.
6. **Scalability** – schema and UI structured to handle increasing order volume during peak hours.
7. **Maintainability** – modular, well-normalized database (3NF/BCNF) and clearly separated UI pages.

2.3. List of AI tools used for building UI(if any):

1.Claude AI

2.4. Prompt (if any) :

Sl. No	Name of the tool	prompts
1	Claude AI	Build a frontend UI for a "Campus Canteen Pre-Order and Queue Management System" using HTML, CSS, and Bootstrap.
2	Claude AI	The system has three user roles: Student (browse menu, place pre-orders, track order status through Placed → Preparing → Ready → Collected, view order history), Canteen Staff (live queue dashboard to update order status, view kitchen capacity), and Admin (manage menu items/pricing/stock, view sales reports like popular items and orders-by-hour).
3	Claude AI	Include a landing page to choose the role. Use a light, attractive, cohesive color scheme, and don't include any real team member names — use generic labels instead.

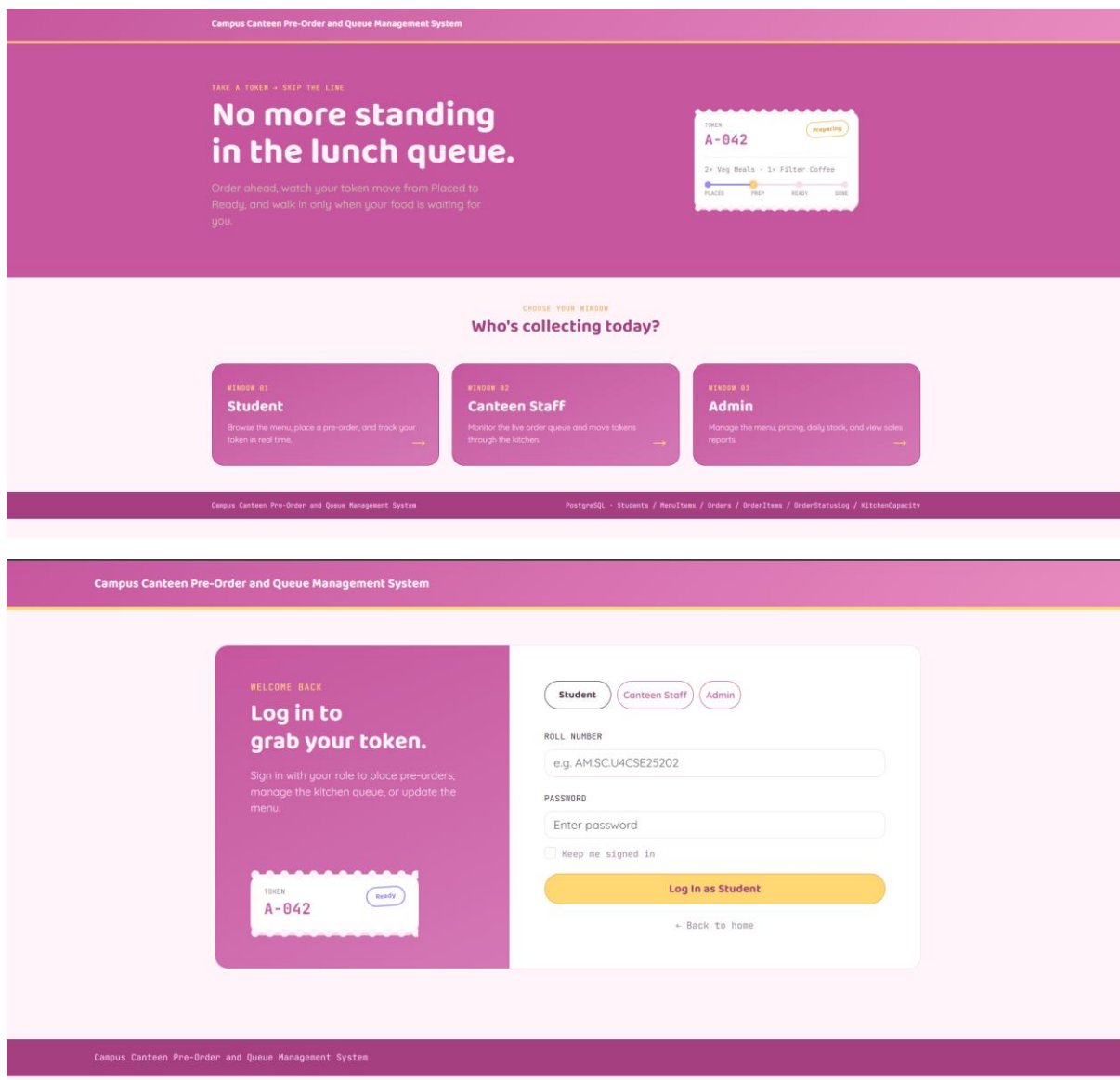
2.5. List of files:

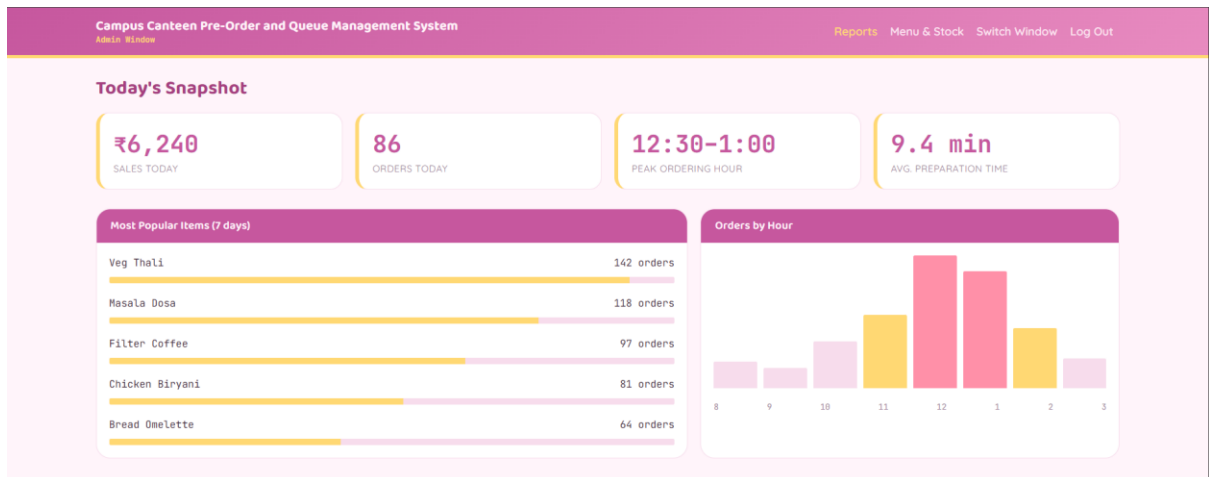
[list the html / JSP/ Java / Python/ any files created for UI]

1. Index.html
2. Student.html
3. Admin.html
4. Staff.html
5. Login.html
6. Styles.css

2.6. UI Screenshot:

[give the screenshot of the UI]

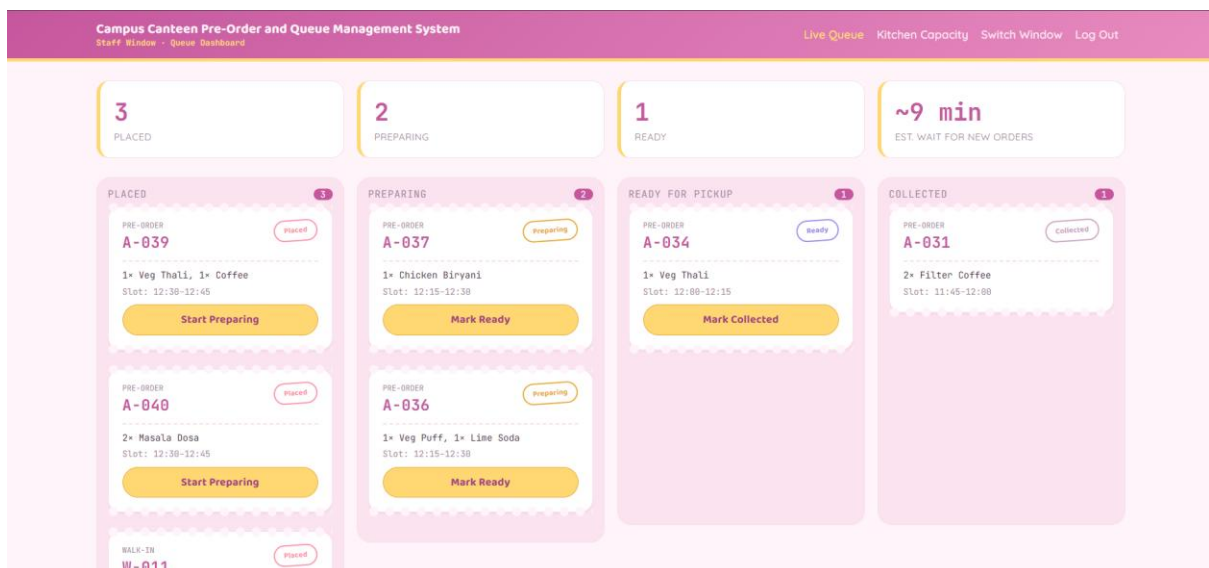


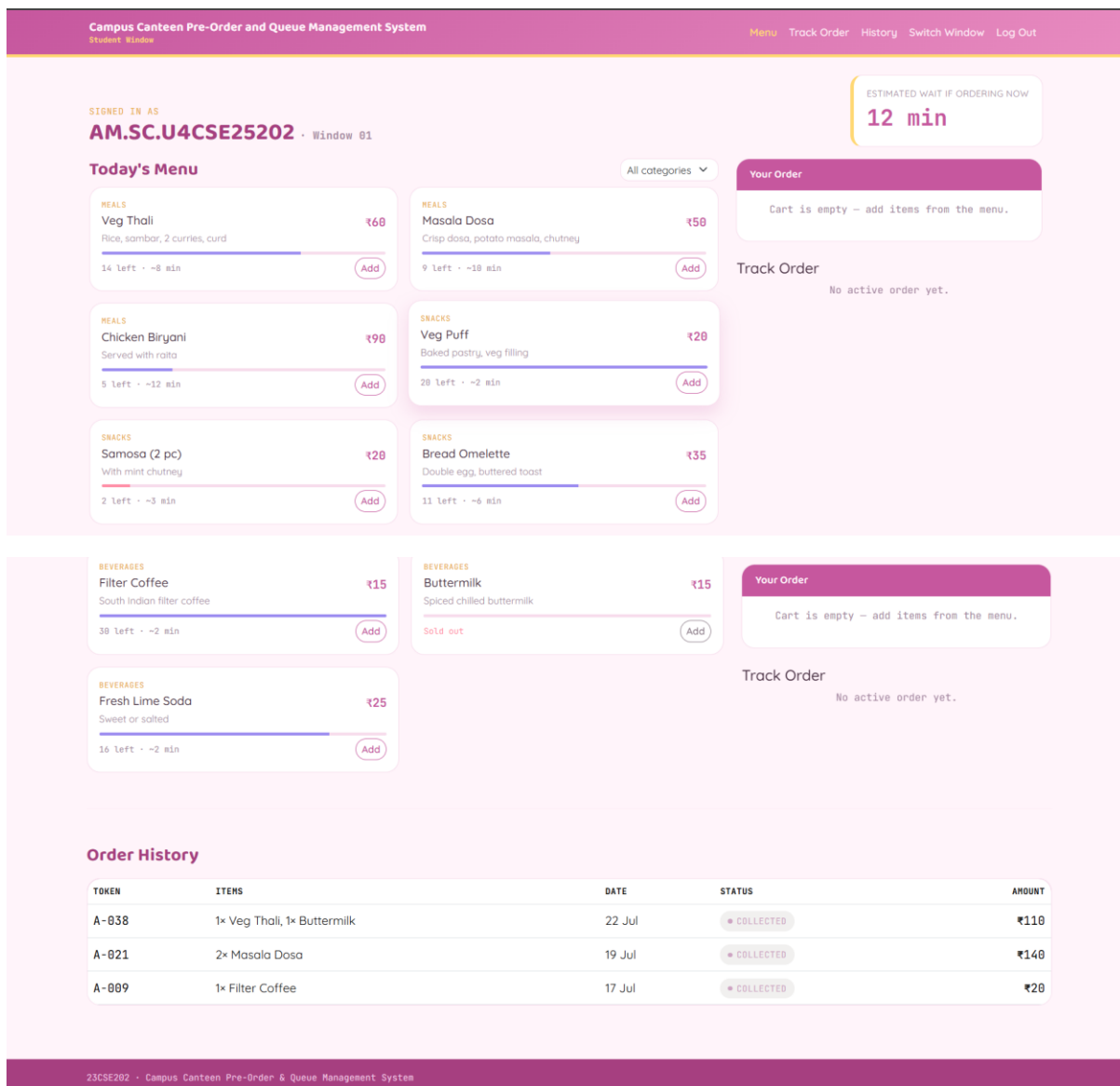
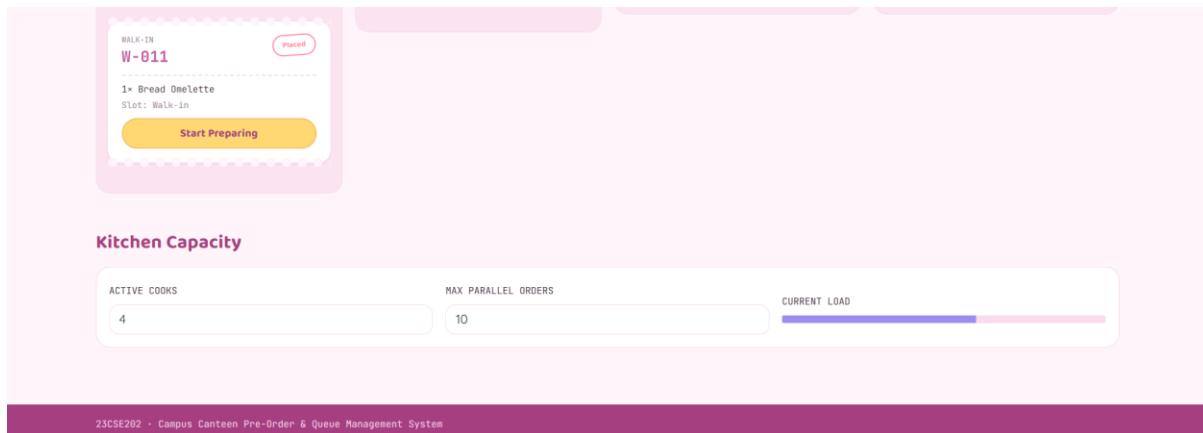


Menu & Stock Management

+ Add Menu Item

ITEM	CATEGORY	PRICE	PREP TIME	STOCK TODAY	STATUS	
Veg Thali	Meals	₹60	8 min	14	AVAILABLE	<button>Edit</button>
Masala Dosa	Meals	₹50	10 min	9	AVAILABLE	<button>Edit</button>
Chicken Biryani	Meals	₹90	12 min	5	AVAILABLE	<button>Edit</button>
Veg Puff	Snacks	₹20	2 min	20	AVAILABLE	<button>Edit</button>
Samosa (2 pc)	Snacks	₹20	3 min	2	AVAILABLE	<button>Edit</button>
Bread Omelette	Snacks	₹35	6 min	11	AVAILABLE	<button>Edit</button>
Filter Coffee	Beverages	₹15	2 min	30	AVAILABLE	<button>Edit</button>
Buttermilk	Beverages	₹15	1 min	0	SOLD OUT	<button>Edit</button>
Fresh Lime Soda	Beverages	₹25	2 min	16	AVAILABLE	<button>Edit</button>





2.7. GitHub Link:

[push the files specified in 2.5 to GitHub repo and provide its link here]

https://github.com/ANAGHA07-GIF/23CSE201_Project